

Bone and Mineral Disorders in Chronic Kidney Disease

Marc Vervloet, Markus Ketteler, Jürgen Floege

DEFINITION

Disturbances of mineral metabolism are ubiquitous in chronic kidney disease (CKD) and lead to serious and debilitating complications unless these abnormalities are addressed and treated. The spectrum of the condition includes abnormal concentrations of serum calcium, phosphate, and magnesium as well as disorders of parathyroid hormone (PTH), fibroblast growth factor-23 (FGF-23), and vitamin D metabolism. These abnormalities and other factors related to the uremic state affect the skeleton and result in the complex disorders of bone known as *renal osteodystrophy*. This term should be used exclusively to define the bone disease associated with CKD, but it is important to acknowledge that aging-related and postmenopausal osteoporosis can exist in parallel. The clinical, biochemical, and imaging abnormalities previously identified as correlates of renal osteodystrophy are now defined more broadly as a syndrome called *chronic kidney disease–mineral and bone disorder* (CKD-MBD).¹ The spectrum of skeletal abnormalities seen in renal osteodystrophy includes (Fig. 88.1):

- Osteitis fibrosa, a manifestation of hyperparathyroidism characterized by increased osteoclast and osteoblast activity, peritrabecular fibrosis, and increased bone turnover.
- Osteomalacia, a manifestation of defective mineralization of newly formed osteoid most often caused by aluminum deposition; bone turnover is decreased.
- Adynamic bone disease (ABD), a condition characterized by abnormally low bone turnover.
- Osteopenia or osteoporosis.
- Combinations of these abnormalities termed *mixed renal osteodystrophy*.
- Other abnormalities with skeletal manifestations (e.g., chronic acidosis, β_2 -microglobulin amyloidosis [A β_2 M amyloidosis]).

EPIDEMIOLOGY

The prevalence of the various types of renal osteodystrophy in patients with end-stage kidney disease (ESKD) is illustrated in Fig. 88.2.² In patients on hemodialysis (HD), osteitis fibrosa is declining, whereas ABD is increasing in prevalence, at least in White patients.³ In Black patients on dialysis, osteitis fibrosa is still the most prevalent form. In patients on peritoneal dialysis (PD), those with diabetes, and the elderly, adynamic bone lesion has been the dominant form of renal osteodystrophy for decades. Osteomalacia represents only a small fraction of cases in either group but is more common in certain ethnic groups, particularly South Asians. Skeletal abnormalities associated with CKD can begin at relatively preserved levels of estimated glomerular filtration rate (eGFR; ~ 50 mL/min/1.73 m²).

PATHOGENESIS

Several biochemical and hormonal abnormalities associated with CKD contribute to renal osteodystrophy and can be affected by efforts at prevention and treatment. The major factors may vary as CKD progresses (Fig. 88.3). Similarly, the predominance of one particular pathogenetic mechanism over another may contribute to the heterogeneity of bone disorders. We therefore discuss separately the two major entities: high- and low-turnover osteodystrophy.

OSTEITIS FIBROSA: HYPERPARATHYROIDISM—HIGH-TURNOVER RENAL BONE DISEASE

Elevated levels of PTH in blood, hyperplasia of the parathyroid glands, and elevations in FGF-23 are seen once eGFR declines below approximately 50 mL/min/1.73 m². Although the level of free (i.e., non-protein bound) calcium in blood is normally the principal determinant of PTH secretion, several metabolic disturbances associated with CKD also alter the secretion of PTH.

Abnormalities of Calcium Metabolism

There are three main body pools of calcium: the bony skeleton (mineral component), the intracellular pool (mostly protein bound), and the extracellular pool (see Chapter 11). The calcium in the extracellular pool is in continuous exchange with that of bone and cells and is altered by diet and excretion. Calcium metabolism mainly depends on the close interaction of two hormone systems, PTH and vitamin D, although a minor and ill-defined role for calcitonin may exist as well. CKD leads to perturbations of both major systems, with adverse consequences on the skeleton. Total serum calcium tends to decrease in parallel with GFR because of phosphate retention, decreased kidney production of 1,25-dihydroxyvitamin D (calcitriol), reduced intestinal calcium absorption, and skeletal resistance to the calcemic action of PTH. However, free calcium levels remain within the normal range in most patients⁴ because of compensatory hyperparathyroidism. Because calcium is a major regulator of PTH secretion, persistent hypocalcemia is a powerful stimulus for development of hyperparathyroidism and also contributes to parathyroid growth.

Abnormalities of Phosphate Metabolism

With progressive CKD, phosphate is retained, at least, transiently, by the failing kidney. However, hyperphosphatemia usually does not become evident before stage 4 CKD. Until then, compensatory hyperparathyroidism and increases in circulating FGF-23 result in increased phosphaturia, maintaining serum phosphate levels in the normal range.⁵

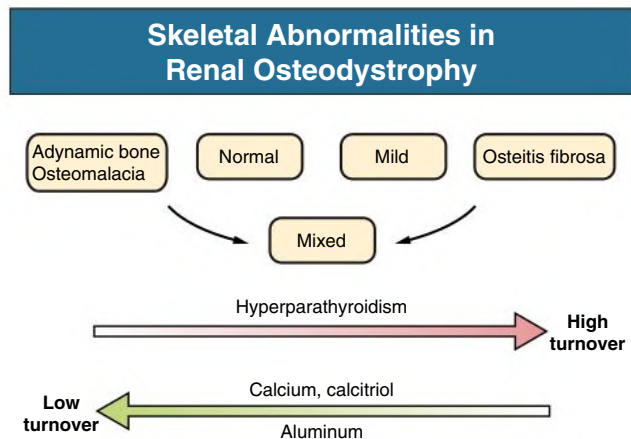


Fig. 88.1 Spectrum of Renal Osteodystrophy. The range of skeletal abnormalities in renal bone disease encompasses syndromes with both high and low bone turnover.

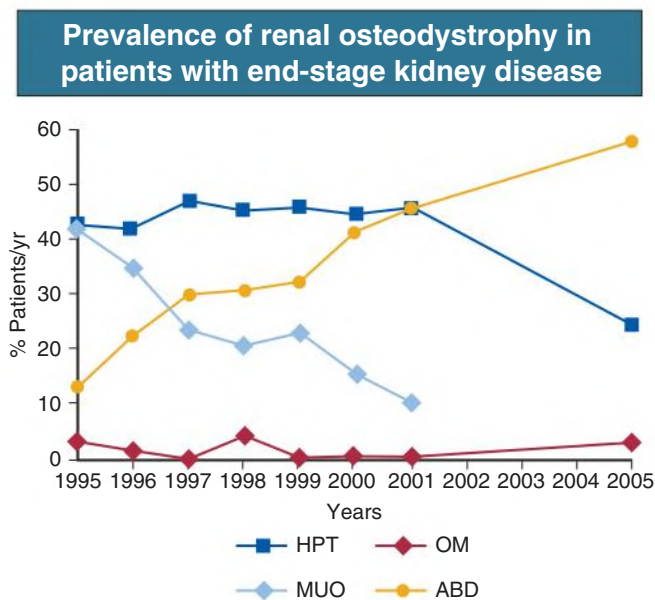


Fig. 88.2 Prevalence of renal osteodystrophy in patients with end-stage kidney disease. ABD, Adynamic bone disease; HPT, high-turnover renal osteodystrophy; MUO, mixed uremic osteodystrophy; OM, osteomalacia. (Modified from Malluche HH, Mawad HW, Monier-Faugere MC. Renal osteodystrophy in the first decade of the new millennium: Analysis of 630 bone biopsies in black and white patients. *J Bone Miner Res.* 2011;26:1368–1376; and Malluche HH, Mawad H, Monier-Faugere MC. The importance of bone health in end-stage renal disease: Out of the frying pan, into the fire? *Nephrol Dial Transplant.* 2004;19[Suppl 1]:i9–13.)

Phosphate retention leads to hyperparathyroidism by decreasing serum free calcium, which, in turn, stimulates the secretion of PTH (Fig. 88.4). In addition, phosphate interacts with the calcium sensing receptors on parathyroid cells, attenuating the suppressive effects of calcium on PTH.⁶ Thus, a new steady state is achieved in which serum phosphate is restored to normal at the expense of a sustained high level of PTH. This cycle is repeated as kidney function declines until sustained and severe hyperparathyroidism is present. Second, phosphate retention leads to decreased kidney production of calcitriol, either directly or by increasing the levels of FGF-23 (which decreases the activity of 1α -hydroxylase). The decrease in calcitriol allows increases in PTH gene transcription and also decreases intestinal calcium absorption, leading to hypocalcemia, which, in turn, also stimulates PTH secretion. Third, hyperphosphatemia

Percentage of Patients Exhibiting Elevated Circulating Levels of Calcium, Phosphate, iPTH, 25OH-Vitamin D₃, Calcitriol, and FGF-23 in Advancing CKD

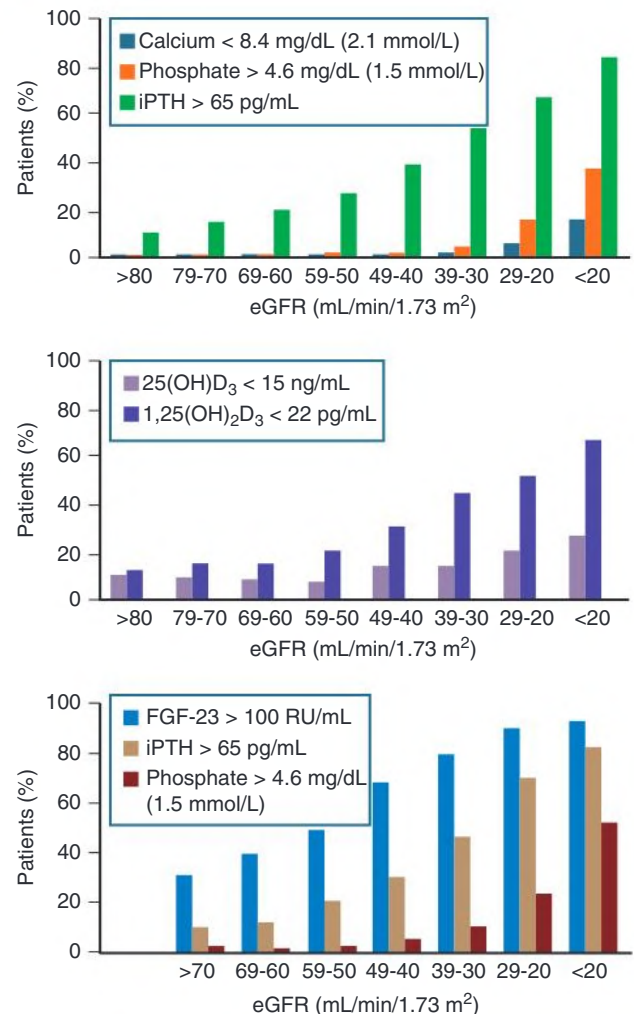


Fig. 88.3 Percentage of patients exhibiting elevated circulating levels of calcium, phosphate, intact parathyroid hormone (iPTH), 25OH-vitamin D₃, calcitriol, and fibroblastic growth factor-23 (FGF-23) in advancing chronic kidney disease (CKD). Particularly in early CKD stages, there is wide variability at the individual level, and some patients, for example, may exhibit increased FGF-23 and normal iPTH, whereas others may have elevated iPTH levels with normal FGF-23 or elevations of both. eGFR, Estimated glomerular filtration rate. (Data from Levin A, Bakris GL, Molitch M, et al. Prevalence of abnormal serum vitamin D, PTH, calcium, and phosphorus in patients with chronic kidney disease: Results of the study to evaluate early kidney disease. *Kidney Int.* 2007;71:31–; and Isakova T, Wahl P, Vargas GS, et al. Fibroblast growth factor 23 is elevated before parathyroid hormone and phosphate in chronic kidney disease. *Kidney Int.* 2011;79:1370–1378.)

is associated with resistance to the actions of calcitriol in the parathyroid glands, which also favors development of hyperparathyroidism and induces resistance to the actions of PTH in bone. Finally, phosphate per se appears to affect PTH secretion independently of changes in serum calcium or serum calcitriol.^{7,8} Phosphate may have an effect on parathyroid growth independent of serum calcium.^{9,10} Regardless of the mechanism by which phosphate retention causes hyperparathyroidism, experimental studies suggest that restriction of dietary phosphate in proportion to the

Phosphate Retention and Secondary Hyperparathyroidism

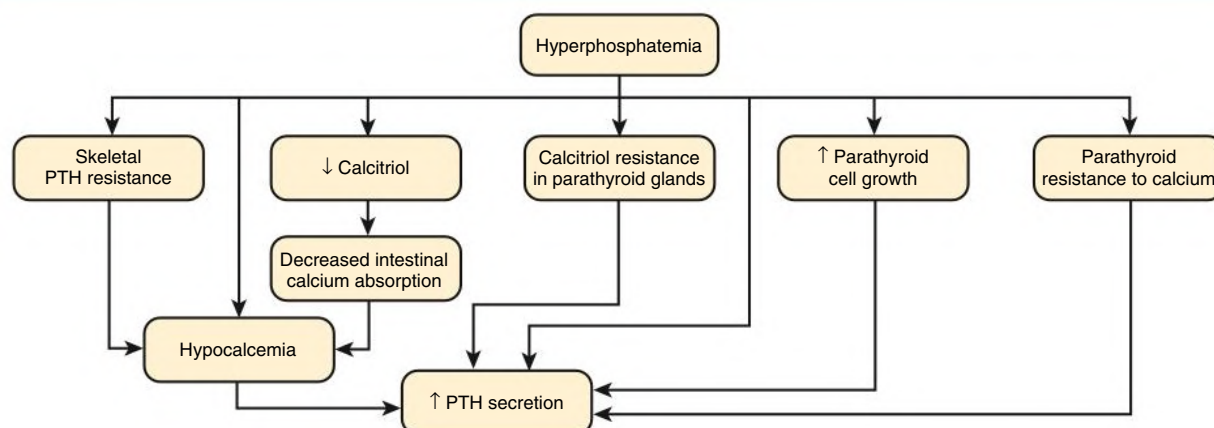


Fig. 88.4 Role of Phosphate Retention in the Pathogenesis of Secondary Hyperparathyroidism. Hyperphosphatemia stimulates parathyroid hormone (PTH) secretion indirectly by inducing hypocalcemia, skeletal resistance to PTH, low levels of calcitriol, and calcitriol resistance. Hyperphosphatemia also has direct effects on the parathyroid gland to increase PTH secretion and parathyroid cell growth.

decrease in GFR can prevent development of hyperparathyroidism, but it is uncertain whether this is clinically relevant. FGF-23 also appears to act directly on the parathyroid gland, with inhibitory effects on PTH secretion and parathyroid growth.^{11,12} This effect is dependent on parathyroid presence of the coreceptor α -klotho, although expression of this coreceptor is reduced in CKD. Elevated FGF-23 levels also lead to a substantial reduction in calcitriol production, which, in turn, contributes to hyperparathyroidism. These various actions may partially explain the association between higher levels of FGF-23 and adverse clinical outcomes.¹³

Abnormalities of Vitamin D Metabolism

The conversion of 25-hydroxyvitamin D to its active metabolite 1,25-dihydroxyvitamin D occurs mainly in the kidney by the enzyme 1α -hydroxylase. Extrarenal production of calcitriol also occurs and contributes to the circulating levels of calcitriol. Kidney calcitriol production progressively declines in parallel with eGFR as a result of several mechanisms (Fig. 88.5).

Calcitriol production is compromised in CKD by a reduction in 25-hydroxyvitamin D levels¹⁴ and the decrease in GFR, which further limits the delivery of 25-hydroxyvitamin D to the site of the 1α -hydroxylase in the proximal tubule. Early increases of FGF-23 in CKD likely have a prominent role on vitamin D metabolism because it inhibits 1α -hydroxylase activity but also promotes catabolism of vitamin D. Phosphate retention, besides increasing FGF-23, may also directly decrease the activity of 1α -hydroxylase. The ensuing reductions in calcitriol levels directly and indirectly contribute to the pathogenesis of hyperparathyroidism as described earlier (Fig. 88.6). Low levels of calcitriol directly release the gene for PTH from suppression by the vitamin D receptor and allow increased PTH secretion. In many tissues, vitamin D regulates its own receptor by positive feedback; in CKD, the vitamin D receptor content is decreased in parathyroid tissue. Administration of calcitriol increases the vitamin D receptor content in the parathyroid glands coincident with the suppression of PTH secretion. Studies in vitro have shown that calcitriol is a negative growth regulator of parathyroid cells and therefore calcitriol deficiency in patients with CKD may facilitate parathyroid cell proliferation. Other direct consequences of low levels of calcitriol contributing to the pathogenesis of secondary hyperparathyroidism include an increase in the parathyroid set-point for calcium-regulated PTH secretion and possibly a decrease in the expression of calcium receptors.

Mechanisms Contributing to Decreased Levels of Calcitriol in CKD

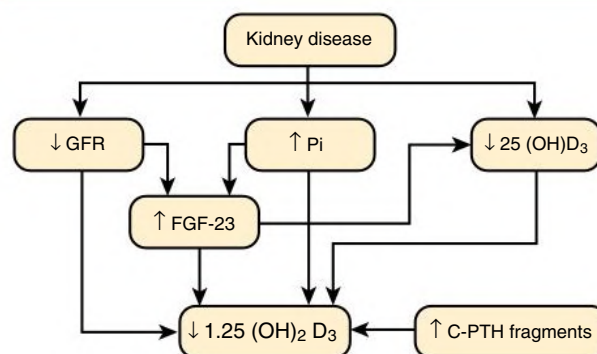


Fig. 88.5 Mechanisms contributing to decreased levels of calcitriol in chronic kidney disease (CKD). C-PTH, Carboxyl-terminal parathyroid hormone; FGF-23, fibroblast growth factor-23; GFR, glomerular filtration rate; Pi, inorganic phosphate.

Low levels of calcitriol also may indirectly promote the development of hyperparathyroidism by reducing intestinal absorption of calcium, leading to hypocalcemia and stimulation of PTH release.

Abnormalities of Parathyroid Gland Function

CKD is associated with intrinsic abnormalities in parathyroid gland function, in addition to those caused by hypocalcemia, low levels of calcitriol, and skeletal resistance to the actions of PTH (Box 88.1).

Parathyroid hyperplasia is an early finding in CKD. In experimental models, hyperplasia begins within a few days after the induction of CKD and can be prevented by dietary phosphate restriction or by the use of calcimimetic agents.^{10,15} Resected parathyroid glands from patients with severe hyperparathyroidism have nodular areas throughout the gland, which represent monoclonal expansions of parathyroid cells. Within these nodules, there is decreased expression of vitamin D receptors, calcium receptors, and α -klotho expression, the cofactor for FGF-23 signaling, reducing the ability of drugs such as active vitamin D and calcimimetics to bind to and treat the gland hyperplasia.

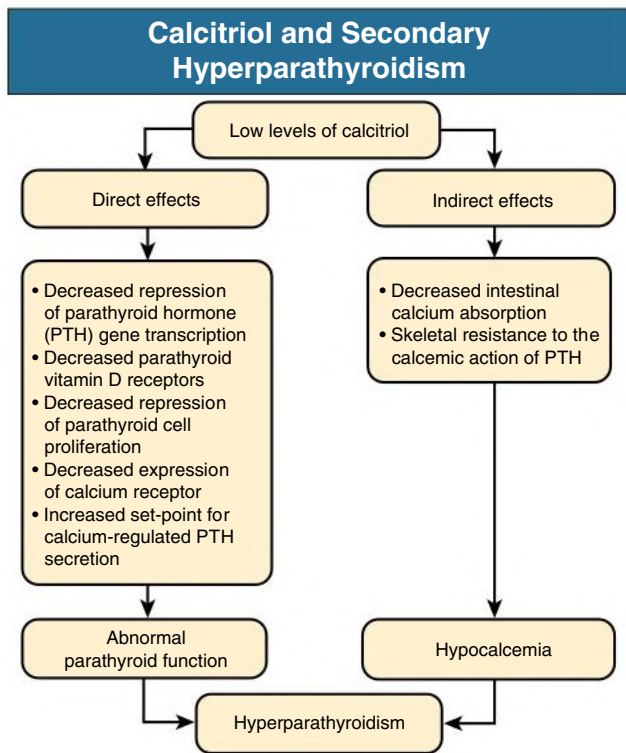


Fig. 88.6 Role of low levels of calcitriol in the pathogenesis of secondary hyperparathyroidism.

The parathyroid calcium receptor is centrally involved in the regulation of PTH secretion by calcium.¹⁶ Its expression and synthesis are decreased in the parathyroid glands from hyperparathyroid patients. In addition, phosphate impairs the calcium sensitivity of the receptor. These mechanisms lead to increased concentrations of calcium to suppress PTH release from the parathyroid cells of uremic patients compared with healthy subjects. Thus, the presence of severe CKD appears to increase the concentration of calcium that is required to substantially decrease PTH release from the parathyroid.

Abnormal Skeletal Response to Parathyroid Hormone

In patients with CKD, there is an impaired response of serum calcium to the administration of PTH and a delay in the recovery from induced hypocalcemia in the presence of larger increments in PTH levels. Thus, in CKD the skeleton is relatively resistant to the calcemic actions of PTH. The resultant decrease in serum calcium levels stimulates PTH secretion and contributes to the pathogenesis of secondary hyperparathyroidism. Factors involved in the skeletal resistance to PTH in CKD include decreased levels of calcitriol, downregulation of the PTH receptor, and phosphate retention. In addition, circulating fragments of PTH, truncated at the N-terminus, have been suggested to oppose the calcemic actions of PTH, possibly acting at a presumed receptor for the C-terminal region of PTH.^{17,18}

Clinical Manifestations of High-Turnover Renal Osteodystrophy

Clinical manifestations of hyperparathyroidism are usually nonspecific and often preceded by biochemical or imaging abnormalities. One specific feature is the increased risk of fractures, typically of long bones, sometimes associated with brown tumors (see later discussion). Nonspecific aches and pains are more common; occur in the lower back, hips, and legs, aggravated by weight bearing; and can be debilitating. Acute, localized bone pain can occur and may be suggestive of acute arthritis. Pain around

BOX 88.1 Parathyroid Abnormalities in Chronic Kidney Disease

- Parathyroid gland hyperplasia: diffuse, nodular
- Decreased expression of vitamin D receptors
- Decreased expression of calcium receptors
- Decreased expression of α -klotho
- Increased set-point of calcium-regulated parathyroid hormone secretion

joints may be caused by acute peri-arthritis, which is associated with peri-articular deposition of calcium phosphate crystals, especially in patients with marked hyperphosphatemia. The symptoms may be confused clinically with gout or pseudogout and often respond to nonsteroidal anti-inflammatory drugs (NSAIDs). The gradual onset of muscle weakness is also common in patients with ESKD. Many factors are probably involved in its pathogenesis, including hyperparathyroidism and abnormalities of vitamin D. $\text{A}\beta_2\text{M}$ (see later discussion) should be considered in the differential diagnosis in very-long-term dialysis patients.

Bone abnormalities may occur in patients with severe hyperparathyroidism, particularly in children, and are manifested on radiographs by subperiosteal erosions, resorption of terminal digits. In adults, deformities arise as a result of fractures. Fractures of the axial skeleton can lead to kyphoscoliosis or chest wall deformities. Slipped epiphysis may occur in children, and frank rachitic features are occasionally evident. Growth retardation is also common in children, and although some improvement has been shown with calcitriol, this is not universal.

Extraskeletal calcifications are frequently encountered in patients with advanced CKD and are aggravated by persistent elevation of the calcium-phosphate product. Most commonly, vascular calcifications are seen, but calcifications may occur in other sites, such as the lung, myocardium, and periarticular areas (Fig. 88.7).

In the skin, hyperparathyroidism can manifest as pruritus (see Chapter 91). Rarely, it can also underlie the development of calciphylaxis (calcific uremic arteriolopathy; see Figs. 91.6 and 91.7).

Diagnosis and Differential Diagnosis

In addition to the clinical manifestations of renal osteodystrophy, a variety of biochemical and radiographic techniques are helpful to establish the specific diagnosis and guide the initiation and adjustment of therapy. Although bone biopsy is not widely used in clinical practice, it remains the gold standard for the diagnosis of renal osteodystrophy and the most specific guide for treatment decisions.

Serum Biochemistry

The levels of free calcium and phosphate in serum are usually normal in patients with mild to moderate CKD. Normally in stage 4 CKD the levels of serum calcium tend to fall and hyperphosphatemia manifests. Hypercalcemia may result from the administration of large doses of calcium-containing antacids or vitamin D metabolites or from severe hyperparathyroid bone disease. It is important to identify the cause of hypercalcemia in patients with CKD (see Chapter 11) because the management will vary greatly according to the cause. Also, the levels of serum calcium and phosphate, when used alone, are not useful in predicting the specific type of bone disease.

Parathyroid Hormone

Measurements of PTH are important for diagnostic purposes and therapeutic guidance in the management of renal osteodystrophy. With kidney impairment, there is accumulation of circulating PTH fragments, which complicates the interpretation of PTH assays, including the two-site immunometric assays, which were thought to measure

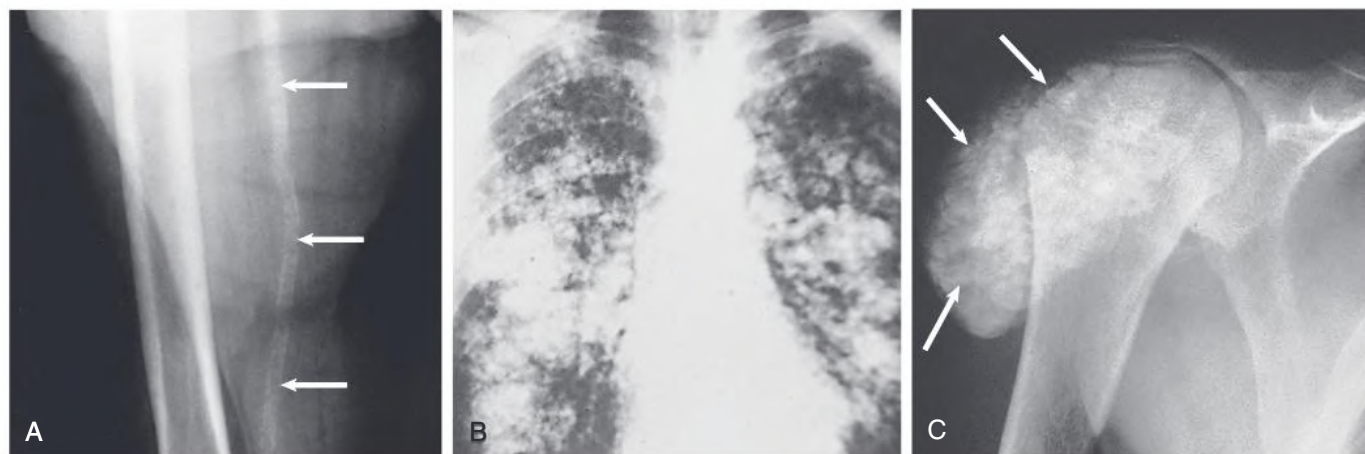


Fig. 88.7 Extraskeletal Calcification in Kidney Failure. (A) Arterial calcification (arrows). (B) Pulmonary calcification. (C) Periarticular calcification (arrows).

“intact” PTH (iPTH). Refinements in PTH assays have demonstrated that these iPTH assays also measure some large fragments of PTH, which are truncated at the *N*-terminus. More specific assays for so-called “biointact” PTH (PTH 1–84) have been developed that exclude these fragments from measurement, but their clinical superiority over iPTH is not established.^{19–21} Standardization needs to be improved among different PTH assays from various laboratories and various manufacturers of assay reagents. With existing iPTH assays (upper limit of the reference range, ~60 pg/mL), only values at the extremes are useful in the noninvasive diagnosis of renal osteodystrophy. In dialysis patients, iPTH levels greater than approximately 600 pg/mL are characteristic of patients with osteitis fibrosa. The threshold values for earlier stages of CKD are not well defined, but there is an element of skeletal resistance to PTH in patients with CKD, and higher levels of PTH promote phosphaturia. Therefore, supranormal levels of PTH appear to be adaptive even in earlier stages of CKD. Serial measurements of PTH are useful in the initial evaluation of patients with renal osteodystrophy and are essential to assess response to therapy and to avoid overtreatment and undertreatment because either can have detrimental effects on bone. There are marked differences among commercial PTH assay results so that precise recommendations of desired ranges cannot reliably be provided and results obtained in different laboratories cannot be easily compared.²²

Vitamin D Metabolites

The levels of calcitriol in patients with CKD are not helpful in differentiating the histologic lesions of renal osteodystrophy. Measurements of calcitriol are not used routinely for diagnostic purposes unless extrarenal production of this metabolite is suspected, as in granulomatous disorders (see [Chapter 11](#)).

Vitamin D deficiency in CKD rarely results in osteomalacia in the United States and Europe but may contribute to hyperparathyroidism. In patients with CKD and marked proteinuria, there is loss of vitamin D-binding protein in the urine, which may result in decreased levels of 25-hydroxyvitamin D. Vitamin D deficiency may be encountered in patients with limited sun exposure, in those with intestinal malabsorption or malnutrition, and in susceptible racial groups, particularly South Asians. Assessment of vitamin D nutrition is by measurement of serum 25-hydroxyvitamin D₃.

Markers of Bone Formation and Bone Resorption

Levels of circulating alkaline phosphatase (AP) offer an approximate index of osteoblast activity in patients with CKD. High levels are

commonly present in hyperparathyroid bone disease. The discriminatory power of AP measurements is enhanced by measurement of γ -glutamyl transferase (γ GT), which lowers the likelihood of the liver of source of elevated AP if γ GT is normal, and bone-specific AP (BAP) isoenzyme, especially in conjunction with PTH values. Serial measurements of AP may be useful in assessing the progression of bone disease. Amino-terminal propeptide of type I procollagen (PINP) may be useful as a biomarker of bone formation, but the level of the monomeric form is dependent on kidney function, which complicates its interpretation in CKD. However, assays detecting trimeric (intact) PINP do not have this limitation. Tartrate-resistant acid phosphatase and collagen degradation products are both markers of osteoclastic activity but are considered investigational at present.

Radiology of the Skeleton

Routine radiographic examination of the skeleton is relatively insensitive for the diagnosis of renal osteodystrophy, and radiographs can appear virtually normal in patients with severe histologic evidence of renal osteodystrophy. However, subperiosteal erosions are often present in severe secondary hyperparathyroidism, detected in the hands ([Fig. 88.8](#)), clavicles, and pelvis. Skull radiographs may show focal radiolucencies and a ground-glass appearance, known as *pepper pot skull*. Osteosclerosis of the vertebrae is responsible for the “rugger-jersey” appearance of the spine ([Fig. 88.9](#)). Very rarely, brown tumors, focal collections of giant cells and typical of severe hyperparathyroidism, are seen as well-demarcated radiolucent zones in long bones, clavicles, and digits ([Fig. 88.10](#)) and may be confused with osteolytic metastases. Looser zones or pseudofractures are characteristic of osteomalacia. Routine skeletal radiographs are not indicated unless there are symptoms. Novel imaging techniques such as magnetic resonance imaging (MRI), positron-emission tomography (PET), and high-resolution peripheral quantitative computed tomography (CT; HR-pQCT) are considered research tools, but some may find their way to clinical practice in the future.

Measurements of Bone Density

Dual-energy x-ray absorptiometry (DEXA) is widely used to assess bone density. Although DEXA does not identify the nature of the underlying osteodystrophy or distinguish this from osteoporosis, it does allow prediction of fracture risk in patients with any stage of CKD, including those after kidney transplantation. Vascular and soft tissue calcifications may contribute to errors in bone density measurements.



Fig. 88.8 Subperiosteal Erosions in Secondary Hyperparathyroidism. Severe subperiosteal erosions as a manifestation of hyperparathyroidism (arrows). The extensive scalloped appearance of the middle phalanx on the left (arrowheads) represents a small brown tumor.



Fig. 88.9 “Rugger-Jersey Spine” in Secondary Hyperparathyroidism. Vertebral bodies show the increased density of the ground plates and central radiolucency, which gives the striped appearance of a rugby jersey.

Bone Biopsy

Biopsy of bone and the microscopic analysis after double tetracycline labeling provide a definitive and quantitative diagnosis of renal osteodystrophy.²³ To standardize reports on bone histology, the Kidney Disease: Improving Global Outcomes (KDIGO) Chronic Kidney Disease–Epidemiology Collaboration (CKD-MBD) work group proposed the TMV classification, an assessment of turnover (T), mineralization (M), and bone volume (V).¹ Bone formation rate is assessed by the administration of two different tetracyclines spaced apart (e.g., tetracycline 500 mg three times daily for 2 days, followed by a 10-day interval, then demeclocycline 300 mg three times daily for 3 days, but several other labeling schemes exist) and biopsy 4 days later. Quantitation of bone mineralization rate is achieved by measuring the distance and surface between the two fluorescent tetracycline bands.

Although noninvasive testing is useful to distinguish low or normal from high bone turnover, there is considerable overlap, and therefore biopsy might be required for definitive diagnosis when biochemistry



Fig. 88.10 Brown tumor (arrow) in a hemodialysis patient with severe hyperparathyroidism. The tumor can easily be confused with a lytic bone metastasis.

is not conclusive (e.g., PTH in recommended range but BAP elevated, hypercalcemia with PTH only modestly elevated, or bone pain), and especially so when treatment decisions depend on bone histology.

Osteitis fibrosa (hyperparathyroid bone disease) is characterized by increased bone turnover, increased number and activity of osteoblasts and osteoclasts, and variable amounts of peritrabecular fibrosis (Fig. 88.11A). Osteoid may be increased but usually has a woven pattern distinct from the normal lamellar appearance. Osteomalacia is characterized by increased osteoid seam width, increase in the trabecular surface covered with osteoid, and decreased bone mineralization as assessed by tetracycline labeling (see Fig. 88.11D). The presence of aluminum can be detected on the mineralization front by specific staining (see Fig. 88.11C). Aluminum-related bone disease is defined by aluminum staining exceeding 15% of the trabecular surface and a bone formation rate of less than 220 mm²/day. Features of osteitis fibrosa may occur together with features of osteomalacia; the combination is termed *mixed renal osteodystrophy*.

Treatment of High-Turnover Bone Disease

Prevention is the primary goal in management of renal osteodystrophy. Therapy for hyperparathyroidism ideally should be initiated in CKD stage 3 so parathyroid gland hyperplasia can be prevented. Because renal osteodystrophy is usually asymptomatic early in the course of CKD, the relevance of secondary hyperparathyroidism is frequently overlooked. By the time CKD is advanced, patients may have already developed significant skeletal abnormalities or nodular parathyroid hyperplasia, and then more aggressive therapy is required to prevent the long-term consequences of renal osteodystrophy. The successful approach to the prevention and management of this disorder involves the integration of a variety of measures directed toward the suppression of PTH secretion and prevention of parathyroid hyperplasia.

Prevention of Hypocalcemia

Hypocalcemia should be corrected because it is a potent stimulus for PTH secretion. Low albumin may mask hypocalcemia unless free calcium is measured. The initial approach to therapy for hypocalcemia in patients with mild to moderate CKD is the administration of calcium supplements such as calcium carbonate, taken between meals with

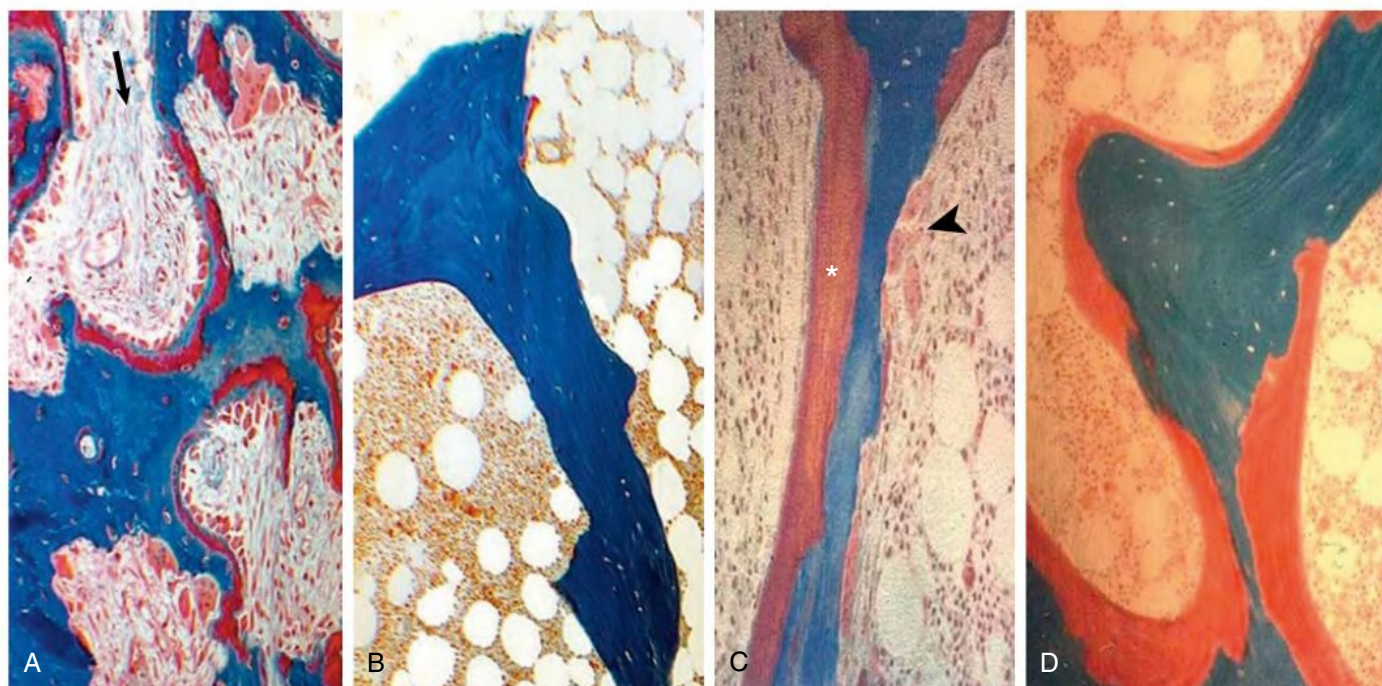


Fig. 88.11 Bone Histology in Renal Osteodystrophy. (A) Osteitis fibrosa: characteristic manifestations of severe hyperparathyroidism with increased multinucleated osteoclasts and osteoblast activity and peritrabecular fibrosis (stained blue, *arrow*). (B) Adynamic bone disease: there is no cellular activity along the bone surface and no osteoid visible. (C) Mixed renal osteodystrophy: there is evidence of active osteoclasts on one bone surface (*arrowhead*) and evidence of thickened osteoid (stained red; *asterisk*) indicating a mineralization defect on the other. (D) Osteomalacia: marked excess of unmineralized osteoid (stained red) surrounding the mineralized bone (stained blue).

increasing doses as required. Vitamin D status should be assessed by measuring 25-hydroxyvitamin D, and deficiency should be corrected if the level is less than 30 ng/mL; 1,25-dihydroxyvitamin D has an 8-hour half-life and so is not helpful for assessing vitamin D status. The efficacy of therapy can be assessed with follow-up determinations of serum calcium and PTH. Adjunctive therapy with active vitamin D sterols should be considered if hyperparathyroidism or hypocalcemia persists. In patients with ESKD, active vitamin D sterols are often required. In dialysis patients, the goal is to achieve levels of iPTH that are approximately two to nine times greater than the upper limit of the assay used.²⁴ Also, iPTH trends over time should be closely monitored. In CKD stages 3 to 5, progressive rises in iPTH above the normal range should be abrogated by correction of hypocalcemia, vitamin D deficiency, and hyperphosphatemia. The latter should be corrected before the correction of hypocalcemia.

Control of Phosphate

Control of phosphate is the cornerstone of effective management of secondary hyperparathyroidism. In mild to moderate CKD, a normal serum phosphate concentration does not necessarily indicate normal parathyroid status, and except for the late stages of CKD, normophosphatemia may be maintained at the expense of elevated serum PTH and FGF-23. Although it is debated whether phosphate-lowering strategies (like dietary restrictions and phosphate binders) are useful if phosphate levels are within the normal range, overt hyperphosphatemia clearly contributes to hyperparathyroidism.

Dietary Phosphate Restriction

Dietary phosphate restriction can prevent excessive PTH synthesis, secretion, and parathyroid cell proliferation in experimental animals

with mild CKD, but the clinical effectiveness of this approach is not proven. Accordingly, restriction of dietary phosphate intake might be considered in patients with CKD stage 2 or 3. The input of a dietician is essential. Protein restriction and limiting dairy products, in particular processed foods containing high amounts of added phosphate, are the mainstays of the regimen. Phosphate-protein restriction increases the serum levels of calcitriol in patients with mild to moderate CKD. However, restriction of phosphate by severe dietary protein restriction below 0.8 g/kg/day may lead to protein-calorie malnutrition.

Phosphate Binders

Although dietary phosphate restriction is usually sufficient in early CKD to treat hyperphosphatemia, this becomes more difficult as kidney function deteriorates. Agents that bind ingested phosphate in the intestinal lumen to limit its absorption are then usually indicated, in addition to dietary restriction. Compounds used for this purpose include calcium-containing antacids; magnesium salts; non-calcium-containing, non-aluminum-containing phosphate binders; and aluminum salts (Fig. 88.12). Several phosphate binders, especially in advanced CKD, have to be given in large numbers (often accounting for 50% of the daily pill burden) and consequently patient adherence with the medication is a major problem, especially because gastrointestinal (GI) side effects (e.g., nausea, vomiting, constipation, diarrhea) occur frequently.

Aluminum-containing antacids are effective phosphate binders, but their long-term use is discouraged in patients with CKD because of the risk for aluminum toxicity. Ingestion of aluminum-containing antacids together with foods containing citric acid (e.g., fruit juices and foods with sodium, calcium, or potassium citrate) may significantly increase aluminum absorption and therefore should be avoided.

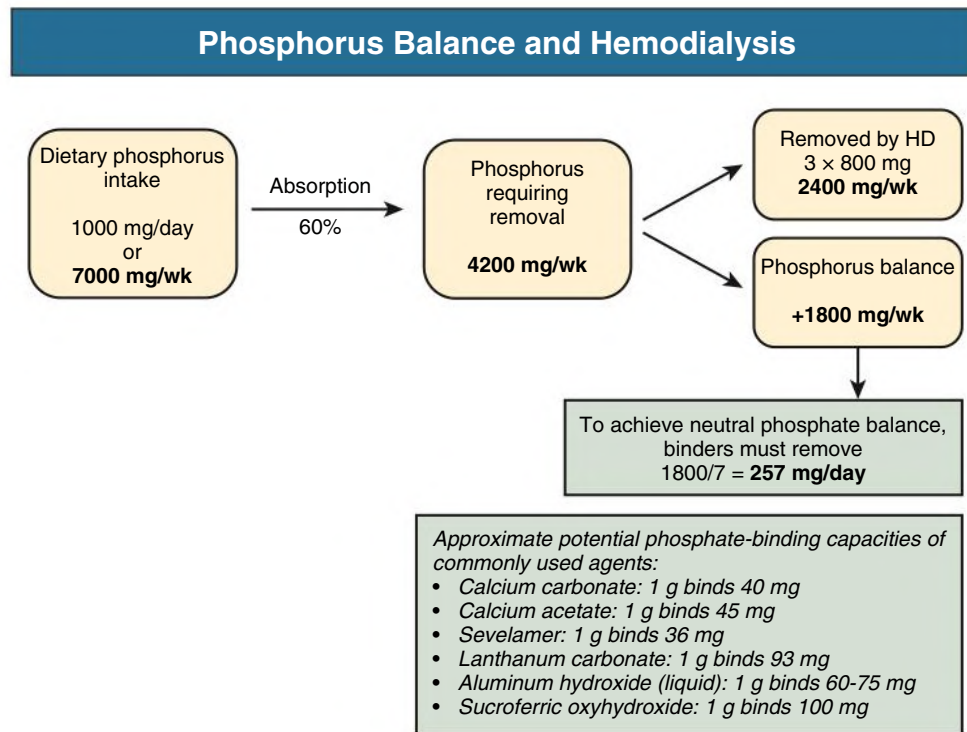


Fig. 88.12 Phosphate balance and phosphate binders used in hemodialysis (HD) patients.

Calcium carbonate or calcium acetate taken with meals effectively binds phosphates and limits their absorption. They are effective phosphate binders in 60% to 70% of patients on HD. The doses required to prevent hyperphosphatemia may vary according to patient compliance with dietary phosphate restriction as well as the CKD stage. Hypercalcemia and calcium loading are potentially serious side effects. Current recommendations are to limit the ingestion of elementary calcium to 1500 mg/day. Consideration of overall calcium balance may be important with the use of calcium-containing phosphate binders.

Magnesium salts are effective phosphate binders for patients who become hypercalcemic with calcium-containing phosphate binders, but they should be administered with caution in CKD patients not on dialysis because hypermagnesemia may have serious adverse effects. In patients on dialysis, magnesium carbonate (elemental magnesium 200–500 mg/day) has been used successfully, with prevention of hypermagnesemia through a reduction in dialysate magnesium concentration, or at least monitoring of magnesium levels.

Sevelamer carbonate is a nonabsorbable, calcium-free polymer, which provides effective phosphate control in a dose range of 2.4 to 4.8 g/day, but higher doses might be needed. Sevelamer may offer an advantage over calcium-containing phosphate binders in terms of limiting the calcium load. The use of sevelamer is associated with decreased progression of vascular calcification.²⁵ Sevelamer may be combined with both calcium- and magnesium-containing phosphate binders if necessary. Lanthanum carbonate also is an effective phosphate binder. No significant toxicity has been observed, and like sevelamer, its use is associated with less progression of vascular calcification compared with calcium-containing phosphate binders.²⁶ In general, pill burden is lower for lanthanum compared with sevelamer. Iron-containing phosphate binders include ferric citrate and sucroferic oxyhydroxide. Both compounds have comparable phosphate binding efficacy as lanthanum per pill. Ferric citrate allows significant oral iron uptake and therapy necessitates monitoring of iron stores, whereas iron uptake from sucroferic oxyhydroxide is low.²⁵

Use of Vitamin D Metabolites

Calcitriol and other 1 α -hydroxylated vitamin D sterols, such as 1 α -hydroxyvitamin D₃ (alfacalcidol), 1 α -hydroxyvitamin D₂ (doxercalciferol), and 19-nor-1 α ,25-dihydroxyvitamin D₂ (paricalcitol), are active forms of vitamin D and are effective for controlling secondary hyperparathyroidism. Calcitriol lowers PTH levels and improves bone histology. In patients with very high levels of PTH and markedly enlarged glands with severe nodular hyperplasia, the effectiveness of vitamin D metabolites may be limited because the expression of vitamin D receptor is low in such tissue. Accordingly, it appears rational to initiate treatment of secondary hyperparathyroidism with active vitamin D early in CKD when the parathyroid glands are more sensitive to such therapy and thereby prevent the progression to a refractory stage. Levels of 1,25 dihydroxycholecalciferol generally correlate with 25 hydroxycholecalciferol. Therefore, generally the first step in treatment of secondary hyperparathyroidism is correction of (native) vitamin D deficiency. A beneficial effect of active vitamin D therapy in treatment of secondary hyperparathyroidism in patients with mild to moderate CKD has been shown, but the concern with initiation of active vitamin D therapy at this stage of CKD is acceleration of the progression of kidney disease, should hypercalcemia occur. Because of the effect of calcitriol to increase intestinal phosphate absorption, hyperphosphatemia and elevations in calcium-phosphate product may predispose patients to the development of metastatic calcification; however, it appears that doses of 1 α -hydroxyvitamin D₃ or calcitriol up to 0.5 μ g/day are not commonly associated with hypercalcemia, hyperphosphatemia, or worsening kidney impairment. Another concern with the use of active vitamin D metabolites before dialysis is that oversuppression of PTH may increase the risk for adynamic bone. Accordingly, vitamin D therapy should be monitored carefully and should not be instituted without documentation of hyperparathyroidism, correction of 25-hydroxyvitamin D deficiency, and prior control of serum phosphate.

In patients with ESKD, indications for therapy with active vitamin D are better defined; however, hypercalcemia and aggravation of

hyperphosphatemia are frequent complications of therapy. In these patients, the PTH suppressive effects of active vitamin D compounds are well established. Several, but not all, observational studies have suggested there may be a survival advantage associated with active vitamin D administration in patients with CKD and with ESKD.^{27,28} However, in a prospective Japanese clinical trial in HD patients with relatively low plasma iPTH levels, oral alfacalcidol therapy did not result in lower mortality or cardiovascular event rates.²⁹ Thus, the principal indication to use active vitamin D is control of secondary hyperparathyroidism.

Role of Calcimimetics

An additional approach to the treatment of hyperparathyroidism in ESKD is the use of calcimimetic agents, such as the oral compounds cinacalcet and evocalcet (approved in Japan), or the intravenously administered etelcalcetide, which targets the calcium-sensing receptor and increases its sensitivity to calcium. In dialysis patients, cinacalcet results in a significant fall in PTH levels and can facilitate the control of hyperparathyroidism. The addition of cinacalcet to standard therapy, which included vitamin D sterols in most participants, in patients with iPTH serum levels exceeding 300 pg/mL while receiving standard therapy allowed significantly more dialysis patients to achieve guideline targets for calcium, phosphate, and iPTH.³⁰ Cinacalcet is especially useful in patients with marginal or frank hypercalcemia or with hyperphosphatemia because it lowers these levels too. In addition, calcimimetics can be used in conjunction with other therapies and may even be more effective when combined with low-dose active vitamin D, which increases the expression of the calcium sensing receptor. Key side effects include hypocalcemia and nausea and vomiting; the latter can be somewhat improved by administering cinacalcet at night and possibly by switching to another calcimimetic. Cinacalcet may attenuate progression of cardiovascular (CV) calcification.³¹ A large clinical trial that examined the effect of cinacalcet compared with placebo on CV events and survival in HD patients with secondary hyperparathyroidism did not show statistical significance of treatment on the primary endpoint. However, this trial had a high risk of missing a beneficial effect of cinacalcet (type II error) because a substantial proportion of the placebo arm initiated commercial cinacalcet and because cinacalcet patients were older on average and thus at higher CV risk than those randomized to placebo.³² Nonetheless, cinacalcet significantly reduced rates of parathyroidectomies and of calciphylaxis compared with placebo.³² In CKD patients not on dialysis, the use of calcimimetics is accompanied by significant phosphate retention and is currently not recommended. The parenterally administered etelcalcetide offers improved patient adherence, but GI adverse events were similar to those with cinacalcet.³³

Role of Parathyroidectomy

Although medical treatments are often effective for the control of hyperparathyroidism, there are patients in whom they are unlikely to be successful, have failed, or are contraindicated and parathyroidectomy should be considered (Box 88.2). Severe hyperphosphatemia in these patients precludes the use of vitamin D because of the risk for metastatic calcification. Some control of iPTH levels may be obtained with calcimimetics, but even these compounds may fail in severe hyperparathyroidism because of downregulated calcium-sensing receptors in the parathyroid glands. However, baseline levels of PTH do not predict response to calcimimetics. Some patients with severe hyperparathyroidism may become hypercalcemic. It is important to be certain that hypercalcemia represents severe hyperparathyroidism and is not caused by adynamic bone or other disease. In some cases, a bone biopsy may be required for a definite diagnosis before proceeding to surgical treatment of hyperparathyroidism. When hypercalcemia in a CKD patient is because of hyperparathyroidism, the level

of iPTH generally exceeds 1000 pg/mL and BAP is usually elevated. Surgical parathyroidectomy might be considered in patients with very severe hyperparathyroidism who may receive a kidney transplant in the near future, particularly if they are female and have significant osteopenia. Parathyroidectomy in these patients can help avoid post-transplantation hypercalcemia and hypophosphatemia (caused by PTH-induced phosphaturia), as well as osteopenia. Avoiding hypercalcemia may lead to improved graft function and possibly to less intra-graft calcification. Less severe hyperparathyroidism may subside spontaneously after kidney transplantation. Parathyroidectomy might be considered in patients with severe hyperparathyroidism who have evidence of metastatic calcification. The development of calciphylaxis is an urgent indication for parathyroidectomy if PTH levels are elevated (see Chapter 91).

The most commonly used surgical procedures are subtotal removal of the parathyroid glands and total removal of the parathyroid glands with reimplantation of parathyroid tissue in the forearm. Recurrence of hyperparathyroidism occurs in about 10% of patients. Total parathyroidectomy alone is less commonly performed because it results in hypoparathyroidism. Total parathyroidectomy with forearm implantation on the brachioradial muscle, or subtotal parathyroidectomy in the neck, marking remaining tissue with clips, may be performed. Unregulated tumor-like growth of parathyroid tissue implants has been described and may be related to the monoclonal nature of the nodular hyperplasia of severe hyperparathyroidism. For patients with forearm implantation, it is important to consider which phlebotomy site is used for PTH levels because local levels may differ from systemic.

Recurrence of hyperparathyroidism may respond to further medical therapy, but repeated surgery to remove the forearm implant or a second neck exploration to search for additional glands is often necessary and may be guided by parathyroid imaging.

Synthesis of Therapeutic Strategies

The general recommendations for the prevention and therapy of renal osteodystrophy are summarized in Fig. 88.13, in which therapeutic maneuvers are stratified according to the stage of CKD.

Intervention should be considered in stage 2 or 3 CKD (GFR, 30–90 mL/min/1.73 m²), and dietary phosphate intake may be restricted once patients enter CKD stage 3. Levels of iPTH should be measured; if elevated above the normal range, the levels of 25-hydroxyvitamin D should be measured and corrected if less than 30 ng/mL. If relevant hyperparathyroidism persists, phosphate binders should be considered if phosphate levels are higher than normal. The effectiveness of calcium-containing phosphate binders in earlier stages of CKD is doubtful. As CKD progresses within stage 3, dietary phosphate restriction should be continued or intensified, with emphasis on inorganic

BOX 88.2 Indications for Parathyroidectomy

- Severe hyperparathyroidism
 - With persistent hyperphosphatemia
 - Unresponsive to calcitriol and calcium
 - With hypercalcemia
 - With intolerance or unresponsiveness to calcimimetics
 - In kidney transplantation candidate if severe and unresponsive to calcitriol and calcimimetics
 - With evidence of metastatic calcification
 - If persistent and clinically relevant >18–24 months after kidney transplantation
- Calciphylaxis with evidence of hyperparathyroidism and no or modest response only to calcimimetic therapy

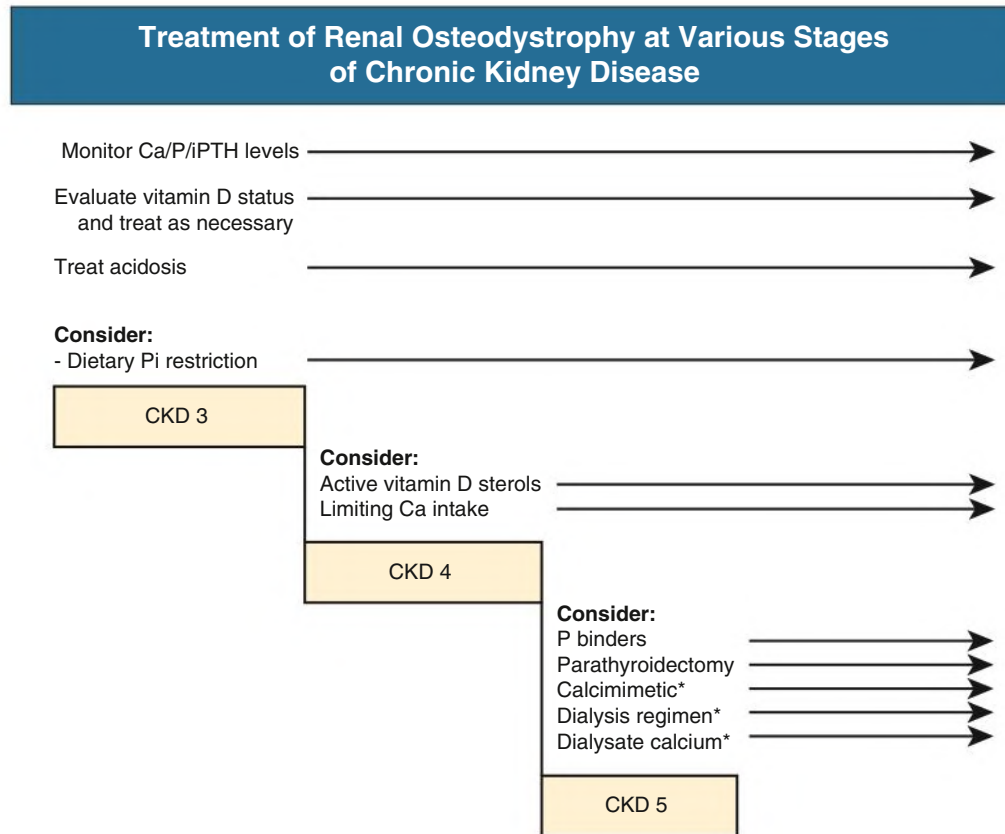


Fig. 88.13 Treatment of renal osteodystrophy at various stages of chronic kidney disease. Asterisk, Consider in CKD stage 5D (i.e., dialysis dependent) only. Ca, Calcium; CKD, chronic kidney disease; iPTH, intact parathyroid hormone; P, phosphate; Pi, inorganic phosphate. (Modified from Martin KJ, Gonzalez EA. Metabolic bone disease in chronic kidney disease. *J Am Soc Nephrol.* 2007;18:875–885.)

phosphate from food additives, and the doses of phosphate binders should be adjusted based on serial measurements of phosphate, with careful attention to avoid hypercalcemia or excessive calcium load. Acidosis, if present, should be treated with oral sodium bicarbonate because persistent acidosis has deleterious effects on the skeleton. The additional sodium load may require further salt restriction or increases in diuretics. Aluminum-based phosphate binders should be avoided. If hyperparathyroidism (iPTH more than two to three times above the upper normal range of the assay) persists despite these measures, consideration should be given to the addition of calcitriol (0.25–0.5 µg/day), active vitamin D analogs, after restoring vitamin D (25OH-vitamin D) deficiency. This therapy should be monitored carefully to avoid hypercalcemia and acceleration of progression of CKD.

In CKD stages 4 and 5, the preceding therapies may need to be intensified and larger amounts of phosphate binders may be required to treat hyperphosphatemia toward normal levels. In patients on dialysis, calcitriol therapy can be intensified, with attention to the serum levels of calcium and phosphate and monitoring of iPTH levels. In CKD stage 5, iPTH levels should be maintained approximately 2 to 9 times above the upper limit of the assay used to maintain normal bone turnover.²⁴ Within this wide range, treatment decisions on PTH management depend on PTH trends and concurrent levels of phosphate, calcium, and BAP.²⁴ Calcitriol or other analogs of active vitamin D may be administered orally either daily or intermittently (pulse therapy) or administered intravenously to patients on HD. During therapy with active vitamin D, it is imperative to ensure serum phosphate remains controlled and elevations of serum calcium are avoided. Cinacalcet provides additional effective control of hyperparathyroidism in patients with ESKD and may be used alone or in combination with the other strategies if iPTH levels

do not fall into the target range. Parathyroidectomy should be considered in selected circumstances. Bone biopsy may be indicated in selected patients, particularly if biomarkers are inconclusive and important treatment decisions are needed (i.e., long-term use of antiresorptive agents or parathyroidectomy), hypercalcemia remains unexplained, or aluminum overload is suspected. Aluminum overload may require chelation therapy with deferoxamine in selected circumstances, especially if it is symptomatic, but in most patients preventing further aluminum exposure will lead to a gradual reduction in the serum levels of aluminum without the need for chelation. During therapy with potent vitamin D compounds, attention should be given to the dialysate calcium concentrations because higher concentrations may aggravate hypercalcemia. However, the increasingly frequent use of lower dialysate calcium levels, such as 1.25 mmol/L, requires careful monitoring of the patient to ensure compliance with calcium-containing phosphate binders and vitamin D metabolites to avoid progressive negative calcium balance and ongoing stimulation of PTH. Dialysate calcium should remain within the range of 1.25 to 1.50 mmol/L and, when possible, should be individually prescribed.²⁴

LOW-TURNOVER (ADYNAMIC) BONE DISEASE

Adynamic bone disease (ABD) describes the morphologic consequences of low-turnover osteopathy in CKD. As CKD progresses, hyperparathyroidism initially develops as an adaptive response to counteract the increasing skeletal PTH resistance and phosphate overload. ABD mainly results from too rigorous suppression of this adaptive response. ABD is increasingly important in CKD-MBD because of the high percentage of affected individuals (>40% in CKD stage 5) and because of its association with CV calcification and mortality.^{34,35}

Pathogenesis of Adynamic Bone Disease

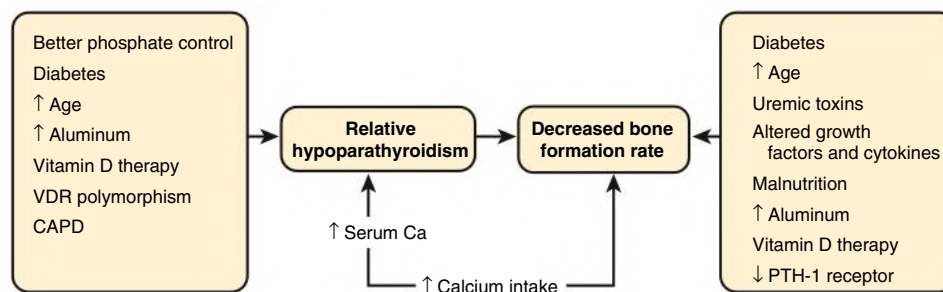


Fig. 88.14 Pathogenesis of adynamic bone disease. Ca, Calcium; CAPD, continuous ambulatory peritoneal dialysis; PTH, parathyroid hormone; VDR, vitamin D receptor. (Modified from Couttenye MM, D'Haese PC, Verschoren WJ, et al. Low bone turnover in patients with renal failure. *Kidney Int Suppl.* 1999;73:S70–S76.)

Furthermore, fracture incidence is estimated to be twice as high in individuals with low than in those with high bone turnover. ABD prevalence is markedly increasing in bone biopsy registries of dialysis patients, which may relate to the increasing prevalence of its key risk factors—advanced age and diabetes mellitus. PD also represents a risk factor for ABD, possibly because of a more continuous exposure to high dialysate calcium as opposed to the cyclic exposure associated with HD.

Pathogenesis of Adynamic Bone Disease

Given that the bone develops a relative resistance of the PTH-1 receptor to its ligand PTH as CKD progresses, PTH levels above the normal range are required to maintain adequate bone turnover. Unfortunately, there are no definite ranges of elevated PTH levels that can reliably differentiate an adaptive response (normal bone turnover) from a maladaptive response (increased bone turnover) because PTH resistance individually varies and because it depends on the stage of CKD. Accordingly, ABD is a consequence of inadequately low PTH levels, which cause suppression or cessation of both osteoblast and osteoclast activities, resulting in a reduced bone formation rate and low bone mass. Iatrogenic oversuppression of PTH in CKD mainly results from high-dose active vitamin D analog treatment, from calcium loading (high doses of calcium-containing phosphate binders, high dialysate calcium concentration), or after parathyroidectomy. The effects of calcimimetic treatment on bone turnover have been prospectively evaluated in HD patients, and patients in this cohort had significant high-turnover osteopathy at baseline (average iPTH > 1200 pg/mL). Therapy with cinacalcet decreased elevated bone formation rate and improved bone histologic status.³⁶ ABD occurred only if iPTH was suppressed below the KDIGO target range of two to nine times the upper reference range of the assay. Finally, diabetes, uremic toxins, malnutrition, and, potentially, C-terminal PTH fragments may be additional factors favoring a state of low bone turnover (Fig. 88.14).

Diagnosis and Differential Diagnosis

Serum Biochemistry

Low iPTH levels (<100–150 pg/mL) are almost always indicative of low bone turnover in CKD stage 5D. However, histologically proven ABD may occur in many dialysis patients with iPTH levels of up to 300 pg/mL and, in exceptions, of up to 600 pg/mL.^{37,38} Therefore, PTH levels alone do not reliably identify ABD. Serum levels or activities of AP or BAP are usually normal or low; downward trends may indicate the development of ABD. Serum calcium and phosphate can be normal or elevated, dependent on the choices of cotreatment (phosphate binders, vitamin D metabolites) and nutritional status. Particularly in instances

Extraskelatal Calcification in Adynamic Bone Disease

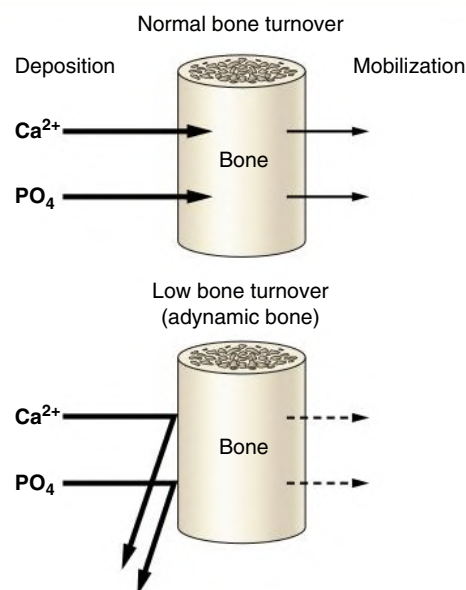


Fig. 88.15 Bone Turnover in Adynamic Bone Disease. Reduced bone turnover leads to increased extraskelatal calcification. Ca^{2+} , Calcium ion; PO_4 , phosphate.

of calcium and phosphate loading, hypercalcemia and hyperphosphatemia may be pronounced because adynamic bone is less able to buffer calcium and phosphate loads by osseous deposition (Fig. 88.15).

In CKD stages 3 and 4, there are uncertainties about the diagnosis of ABD and its clinical consequences. It is unclear which PTH levels are required to maintain adequate bone turnover in these stages. It seems reasonable to correct vitamin D deficiency, hyperphosphatemia, and hypocalcemia when PTH levels start to rise, but beyond that, no firm recommendations can be given.

Bone Biopsy

The gold standard for diagnosis of ABD is bone biopsy. According to the TMV classification (see earlier discussion),¹ ABD is characterized by low turnover, normal (or high secondary) mineralization, and low bone (osteoid) volume. The individual indication to perform bone biopsy should be considered in symptomatic patients based on inconsistencies of biochemical parameters associated with unexplained

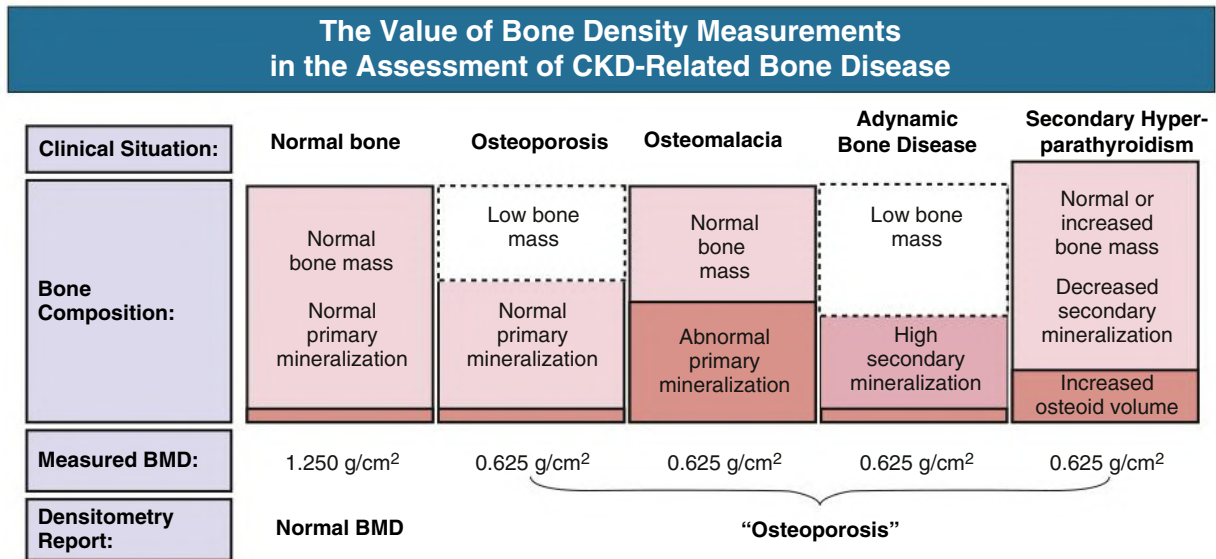


Fig. 88.16 Value of bone density measurements in the assessment of chronic kidney disease (CKD)-related bone disease. Pink boxes indicate mineralized bone; red boxes indicate osteoid. BMD, Bone mineral density. (Courtesy Prof. M.H. Lafage-Proust, St. Etienne, France.)

fractures, bone pain, progressive extraosseous calcifications, or hypercalcemia. The KDIGO initiative recently investigated more than 492 bone biopsy samples and concomitant serum samples to identify biomarker patterns that may allow noninvasive assessment of bone turnover status. ABD was quite strongly associated with low-normal to low BAP levels. In general, PTH and BAP could distinguish low from non-low and high from non-high bone formation rate.³⁹ However, there is considerable heterogeneity between the extremes. Aluminum toxicity is a relevant differential diagnosis versus ABD. Thus, if the patient's history suggests significant aluminum exposure, aluminum bone deposition should be excluded by measurement of serum aluminum and specific staining of a bone biopsy specimen.

Radiology and Measurements of Bone Density

There are no typical features of ABD in conventional bone radiographs or DEXA. In the latter, bone density may be low, normal, or high, depending on the primary or secondary mineralization state, but it never reflects the actual turnover and is therefore not a helpful diagnostic test (Fig. 88.16). A very high CV calcification burden on conventional radiographs may raise the suspicion of a low bone turnover state, if accompanying biochemical parameters are compatible with this diagnosis. Biopsy-proven ABD is associated with the highest magnitude of vascular calcification in dialysis patients.³⁴

Treatment of Adynamic Bone Disease

The key therapeutic approaches in the treatment of ABD are to avoid PTH overexpression and restore adequate PTH levels, without triggering the progressive development of secondary hyperparathyroidism. Such a stepwise treatment approach may include reduction or avoidance of calcimimetics, reduction or avoidance of active vitamin D metabolites, reduction or avoidance of calcium-containing phosphate binders, and reduction of the dialysate calcium concentration (usually to 1.25 mmol/L), first of all based on the magnitude and development of serum PTH levels. Indeed, a recent randomized controlled trial found a histologic improvement of low bone turnover when the dialysate calcium concentrations were decreased from 1.5 or 1.75 mmol/L to 1.25 mmol/L.⁴⁰ Any aluminum should be withdrawn. After these interventions, biochemical parameters (PTH, calcium, phosphate, perhaps AP, and especially BAP) should be monitored more frequently than usual.

Studies comparing calcium-containing with non-calcium-containing phosphate binders in dialysis patients^{38,41} found that the administration of calcium-containing phosphate binders was associated with a higher proportion of individuals who developed ABD. This development was associated with a fall in serum PTH because of the higher calcium load. In an observational study, high-dose calcium-containing phosphate binder intake was associated with both low bone turnover and increased aortic calcification.⁴²

Other therapeutic approaches to ABD have not been systematically studied. They include optimized diabetes control, a change from PD to HD to facilitate a more flexible dialysate calcium prescription, and calcilytics (agents that antagonize the calcium receptor and thus increase endogenous PTH).⁴³ At present, many patients with ABD remain refractory to these treatments.

Future options for treatment might include recombinant PTH (teriparatide, 1–84 PTH), especially in cases of "irreversible" ABD (i.e., after total parathyroidectomy). Supportive study data on relevant clinical features (e.g., bone histomorphometry, bone mineral density, hypercalcemia) are becoming increasingly available from case reports and non-CKD populations,^{44–46} but this approach would be considered an "off-label" treatment for ABD.

OSTEOPOROSIS IN CHRONIC KIDNEY DISEASE

Although abnormal bone is common and fracture risk is increased in CKD patients, the relative contribution of classic osteoporosis (as defined by World Health Organization [WHO] criteria) to the CKD-MBD complex is not well defined. Data from studies of antiosteoporosis agents are mostly available for patients in CKD stages 1 to 3, and subjects with features of CKD-MBD were largely excluded. Nevertheless, postmenopausal women and elderly men are highly prevalent in late-stage CKD populations, and it is thus likely that classic osteoporosis also contributes to their bone disease.

Pathogenesis of Osteoporosis in Chronic Kidney Disease

Osteoporosis may be associated with low, normal, or high bone turnover and is characterized by thin and disconnected trabeculae and loss of the plate-like bone structure. Many patients with CKD have abnormal mineralization and increased osteoid, which is quite atypical for osteoporosis.

Typical pathogenetic factors of osteoporosis, including older age, low estrogen levels, immobilization, and corticosteroid use, are frequent in CKD patients, although some postmenopausal women with late-stage CKD may have relatively normal estrogen levels. However, secondary hyperparathyroidism, relative hypoparathyroidism (as in ABD), and 25-hydroxyvitamin D, as well as 1,25-dihydroxyvitamin D deficiencies, may dominate and “override” the bone phenotype of osteoporosis even if classic risk factors are present.

Diagnosis and Differential Diagnosis

Osteoporosis diagnosis relies on an areal bone mineral density (BMD) evaluated by DEXA at the spine or hip below -2.5 standard deviation (SD) from the BMD in young women (T-score). BMD as assessed by DEXA predicts fractures in patients with CKD G4 to G5D but likely also underestimates the actual risk in patients with CKD G4 to G5D because it does not account for impaired bone quality.⁴⁷ Furthermore, in advanced CKD, BMD measurements do not differentiate classic osteoporosis versus other CKD-MBD-related bone disease, and low BMD can be found in CKD-MBD-induced high-turnover bone disease, ABD, and osteomalacia (see Fig. 88.16). The value of biomarkers for diagnosing osteoporosis in advanced CKD is not well established; if monitoring is attempted, analytes should be preferred that do not exhibit major kidney retention, such as BAP, intact procollagen type I N-propeptide (PINP; marker of bone collagen synthesis), and tartrate-resistant acid phosphatase 5b (TRAP5b; marker of osteoclast activity). Peripheral quantitative tomography (pQCT) of the radius may be a superior methodology for assessment of CKD patients in the future but awaits validation in sufficiently large patient cohorts.⁴⁸ The only reliable methodology to diagnose osteoporosis and discriminate it from other bone manifestations in CKD patients is bone biopsy. In a large bone biopsy study, including 1429 samples from dialysis patients, osteoporosis was diagnosed in 52% of individuals, and 49% of them also had ABD.⁴⁹ These proportions may be quite different in patients in earlier CKD stages, but there are no systematic data available on such cohorts.

Treatment of Osteoporosis in Chronic Kidney Disease

Nonpharmacologic approaches to osteoporosis in CKD (i.e., sufficient calcium supply, exercise, fall prevention) do not differ from the general population. Post-hoc analyses indicate that antiosteoporotic medications are safe and efficacious for treatment of postmenopausal women in stages CKD 1 to 3 and high risk for fractures (according to WHO criteria or using the Fracture Risk Assessment [FRAX] tool) and no features of CKD-MBD.^{50–53} In such populations, bisphosphonates, denosumab, raloxifene, and teriparatide appear to be feasible therapeutic options. The former three drugs antagonize high bone turnover with an antiresorptive mode of action; teriparatide exerts bone anabolic effects (PTH analog). Bisphosphonates may lead to oversuppression of osteoclasts, but only in patients with advanced CKD. A recent subanalysis of patients in CKD stages 2 to 4 from the Following Rehabilitation, Economics and Everyday-Dialysis Outcomes Measurements (FREEDOM) trial demonstrated a significantly reduced fracture risk with denosumab treatment versus placebo, independent of the stage of CKD.⁵⁴ In CKD patients with ABD, bisphosphonates and denosumab may aggravate osteoclast paralysis. In CKD patients with secondary hyperparathyroidism, these antiresorptive agents may upregulate PTH secretion. Patients with advanced CKD (stages 4 and 5) may develop particularly severe hypocalcemia when treated with denosumab; coadministration of vitamin D analogs and calcium may be required to blunt this effect. However, because antiresorptive therapies may be effective in patients with advanced CKD,⁴⁷ and the lack of definite evidence that these drugs induce ABD in this population, it may not be necessary to perform a bone biopsy before treatment

initiation, if bone turnover markers and repeat DEXA scans indicate a clinical response.

β_2 -MICROGLOBULIN-DERIVED AMYLOID

β_2 M amyloidosis, also termed *dialysis-associated amyloidosis*, exclusively and rarely affects patients with long-standing stage 5 CKD. Although the process is systemic, clinical manifestations are largely confined to the musculoskeletal system. β_2 M amyloidosis in CKD stage 5 should not be confused with a rare hereditary systemic amyloidosis derived from the Asp76Asn variant of β_2 -microglobulin, which manifests in the absence of CKD.

Pathogenesis

Fibrils of β_2 M amyloid are derived from the circulating precursor protein β_2 -microglobulin, the nonvariable light chain of the human leukocyte antigen (HLA) class I complex. The pathogenesis appears to involve three events:

1. Pronounced kidney retention of β_2 -microglobulin (11.8 kDa), leading to plasma levels that can be elevated up to 60-fold in dialysis patients⁵⁵; however, even massive overproduction of β_2 -microglobulin in mice was not sufficient to induce amyloid deposits and thus further steps must be important.⁵⁶
2. Modifications of the β_2 -microglobulin molecule that render it more amyloidogenic.⁵⁷
3. Local factors that contribute to and determine the spatial localization of the amyloidosis.

Epidemiology

Histologic studies from the 1990s observed amyloid deposits in 100% of dialysis patients treated for more than 13 years.⁵⁸ Most amyloid deposits never cause clinical problems. β_2 M amyloid-related symptoms presently are largely confined to patients who have been dialyzed for more than 15 years.

Clinical Manifestations and Diagnosis

β_2 M amyloidosis mainly manifests at osteoarticular sites, particularly synovial membranes; visceral manifestations are rare.⁵⁵ Carpal tunnel syndrome is often bilateral and usually requires surgery. Osteoarthropathy of peripheral joints, resulting from amyloid deposition in periarticular bone and the synovial capsule (Fig. 88.17), is

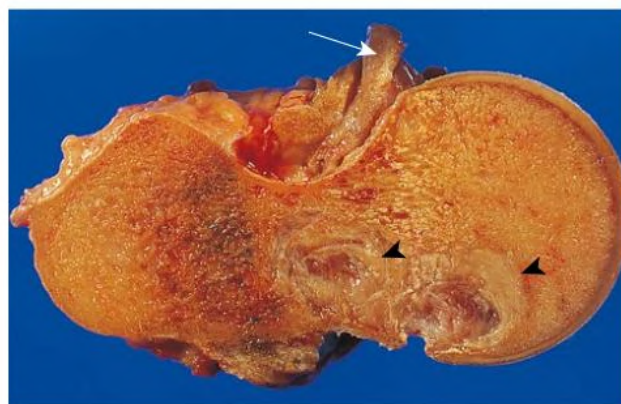


Fig. 88.17 β_2 -Microglobulin (β_2 M) Amyloid Deposition in the Femoral Head. Postmortem specimen from a patient who had undergone long-term hemodialysis. Two large lesions (arrowheads), partly filled with grayish amyloid and partly cystic, are noted in the femoral head. Also note the marked thickening of the synovial capsule from amyloid deposition (arrow).

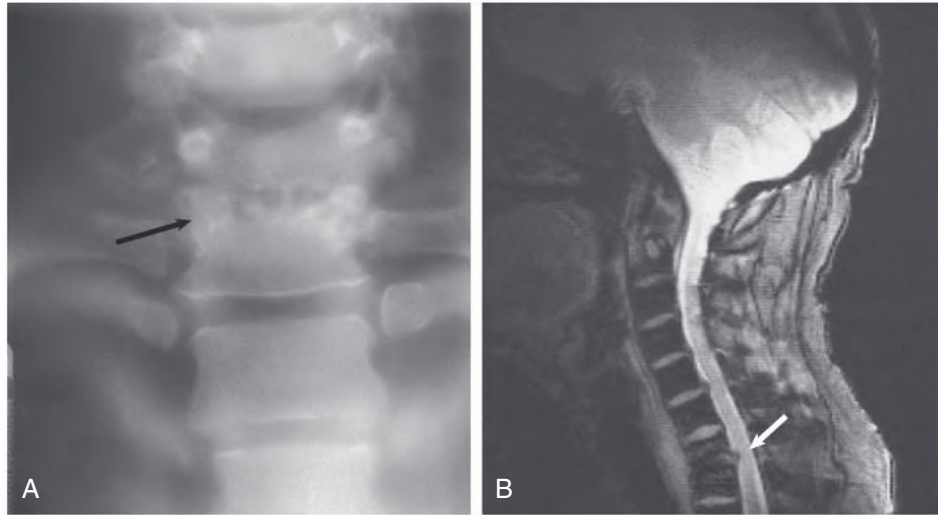


Fig. 88.18 β_2 -Microglobulin ($A\beta_2M$) Amyloidosis–Associated Spondyloarthropathy. (A) Destruction of an intervertebral disk (arrow) in the neck vertebrae of a long-term hemodialysis patient. (B) Magnetic resonance image of the same patient. Note destruction of the intervertebral space and protrusion of material into the spinal canal (arrow).

characterized by recurrent or persistent arthralgias, stiffness of large and medium-sized joints, and swelling of capsules and adjacent tendons. Recurrent joint effusions and synovitis, often in the shoulders and knees, may occur. The clinical presentation may vary from frank, acute arthritis to slow, progressive destruction of the affected joints. Destructive spondyloarthropathy (Fig. 88.18) resulting from $A\beta_2M$ amyloidosis can manifest as asymptomatic deposits, radiculopathy, stiffness, “mechanical ache,” and, finally, medullary compression with resulting paraplegia or cauda equina syndrome. Other manifestations include camptodactyly (a flexion deformity resulting in bent fingers that cannot completely extend or straighten) resulting from amyloid deposits along the flexor tendons of the hands (Fig. 88.19). Patients undergoing dialysis also can have subcutaneous tumorous deposits of $A\beta_2M$ amyloid.

Clinically relevant organ manifestations in patients treated with HD for more than 15 years have described heart failure, odynophagia (painful swallowing), intestinal perforation of both small and large bowel, GI bleeding and pseudo-obstruction, gastric dilation, paralytic ileus, persistent diarrhea, macroglossia or functional tongue disturbances (abnormal taste, mobility, articulation), ureteral stenosis, and nephrolithiasis.

Diagnosis

Serum levels of β_2M do not distinguish between patients with amyloidosis and those without. Ultrasound can detect synovial $A\beta_2M$ amyloidosis as thickening of the joint capsules and tendons and the presence of echogenic structures between muscle groups and joint effusions.^{55,57} On radiologic examination, affected joints may present with single or multiple juxta-articular, “cystic” (i.e., amyloid-filled) bone radiolucencies (Fig. 88.20; see also Fig. 88.17). Such bone defects are prone to pathologic fractures. Diagnostic criteria for $A\beta_2M$ amyloid-induced cystic bone radiolucencies include (1) diameter of lesions more than 5 mm in wrists and more than 10 mm in shoulders and hips, (2) normal joint space adjacent to the bone defect, (3) exclusion of small subchondral cysts in the immediate weight-bearing area of the joint and of defects of the “synovial inclusion” type, (4) increase of defect diameter of more than 30% per year, and (5) presence of defects in at least two joints.⁵⁹ Scintigraphy, using either radiolabeled serum amyloid P component or β_2 -microglobulin,⁶⁰ offers more specific detection of



Fig. 88.19 Hand Involvement in β_2 -Microglobulin ($A\beta_2M$) Amyloidosis. Hand of a long-term hemodialysis patient showing maximal extension. Note the prominence of shrunken flexor tendons (arrows). This is also known as the *guitar string* sign.

amyloid deposits but is not widely available. The definitive diagnosis of $A\beta_2M$ amyloidosis relies on histologic findings. Fat aspiration and rectal biopsy are not helpful in $A\beta_2M$ amyloidosis, but diagnostic material can be obtained from synovial membranes, synovial fluid, or bone lesions.

Treatment and Prevention

Therapy for established $A\beta_2M$ amyloidosis is symptomatic. NSAIDs and physical and surgical measures, such as carpal tunnel decompression, endoscopic coracoacromial ligament release, and bone stabilization in areas of cystic destruction, are all used.⁵⁵ Dialytic removal of β_2M is probably of no therapeutic value in established $A\beta_2M$ amyloidosis. Kidney transplantation is the preferred treatment because it leads to rapid symptomatic improvement and halts further progress of the disease, but whether this can actually lead to regression of established $A\beta_2M$ amyloid deposits is uncertain.

A number of strategies exist for prevention of the clinical manifestations of $A\beta_2M$ amyloidosis.⁵⁵ The risk for carpal tunnel syndrome is reduced by 40% to 50% in patients treated with high-flux hemo(dia)



Fig. 88.20 Peripheral Bone Cystic Radiolucencies in β_2 -Microglobulin ($A\beta_2M$) Amyloidosis. Radiographic findings in a long-term hemodialysis patient. (A) Multiple cystic lesions (*arrows*) are present in the hand bones. (B) Large cysts (*arrows*) in the neck of the femur and adjacent pelvic bones. (C–D) Anterior and lateral views of the head of the tibia with two very large, cystic lesions (*arrows*) resulting in posterior bulging of the tibial plateau.

filtration and minimal in patients receiving online hemodiafiltration. A dramatic reduction in the prevalence of carpal tunnel syndrome occurred in patients dialyzed with ultrapure dialysate. In another

study, an 80% reduction of amyloid signs in a chronic HD population appeared to relate to dialysate factors such as microbiologic purity and the use of bicarbonate buffer.

SELF-ASSESSMENT QUESTIONS

1. A 56-year-old White woman with ESKD secondary to presumed hypertensive nephrosclerosis has been on maintenance HD for 3.5 years. She reports to dialysis treatments regularly and is being adequately dialyzed. Laboratory data over the past few months reveal that intact PTH levels have ranged from 360 to 545 pg/mL, serum calcium 9.5 to 10.5 mg/dL (2.4–2.6 mmol/L), and phosphorus 5.5 to 7.0 mg/dL (1.83–2.33 mmol/L). She has been treated with increasing doses of paricalcitol, dietary phosphate restriction, and calcium carbonate as a phosphate binder. A few days ago, she was seen by her internist regarding vague symptoms that included abdominal pain. At that time, blood tests showed serum phosphorus 7.6 mg/dL (2.5 mmol/L) and calcium 10.1 mg/dL (2.5 mmol/L). The rest of the metabolic profile was within normal limits. A kidney, ureter, and bladder (KUB) film of the abdomen showed calcification of the abdominal aorta. What is the most appropriate next step in the management of this patient's CKD-MBD?
 - A. Increase the dose of paricalcitol.
 - B. Increase the dose of calcium carbonate to improve phosphate control.
 - C. Stop calcium carbonate and add a non-calcium-containing phosphate binder.
 - D. Refer for parathyroidectomy.
 - E. Decrease dialysate calcium.
2. A 75-year-old White man on HD because of diabetic nephropathy exhibits rapid progression of vascular calcification. His total serum calcium fluctuates around the upper limit of normal, with occasional episodes of mild hypercalcemia. Serum phosphorus is around 7 mg/dL (2.3 mmol/L) despite various changes of his phosphate binder medication. The latest intact PTH level is 105 pg/mL (previous levels ranged from 90–180 pg/mL). Current medication includes fluvastatin, calcium acetate, low-dose aspirin, a β -blocker, and insulin. What is the most appropriate next step in the management of this patient's CKD-MBD?

- A. Continue the described therapy.
 - B. Increase the dose of calcium acetate to improve phosphate control.
 - C. Repeat dietary counseling.
 - D. Add low-dose calcitriol (e.g., 0.25 µg/day).
 - E. Replace calcium acetate with a calcium-free phosphate binder.
3. A 68-year-old White woman with an eGFR of 25 mL/min/1.73 m² as a result of presumed hypertensive nephrosclerosis presents for a regular checkup. She has no specific complaints. Medication consists of an ACE inhibitor, a loop diuretic, a β-blocker, and low-dose

aspirin. Current laboratory values include intact PTH level 90 pg/mL (reference range <60), serum calcium 8.1 mg/dL (2.0 mmol/L) and phosphorus 5.7 mg/dL (1.8 mmol/L), 25(OH)-vitamin D3 17 µg/L (target range 30–100). What step in the management of this patient's CKD-MBD is *not* appropriate?

- A. Educate the patient about phosphate content in food.
- B. Initiate therapy with a phosphate binder.
- C. Initiate substitution of 20,000 IU vitamin D3 every second week.
- D. Educate the patient about food additives.
- E. Initiate therapy with a low dose of a calcimimetic.

REFERENCES

- Moe S, Drueke T, Cunningham J, et al. Improving global O. Definition, evaluation, and classification of renal osteodystrophy: a position statement from Kidney Disease: Improving Global Outcomes (KDIGO). *Kidney Int.* 2006;69:1945–1953.
- Sherrard DJ, Hercz G, Pei Y, et al. The spectrum of bone disease in end-stage renal failure—an evolving disorder. *Kidney Int.* 1993;43:436–442.
- Malluche HH, Mawad HW, Monier-Faugere MC. Renal osteodystrophy in the first decade of the new millennium: analysis of 630 bone biopsies in black and white patients. *J Bone Miner Res.* 2011;26:1368–1376.
- Martinez I, Saracho R, Montenegro J, et al. The importance of dietary calcium and phosphorous in the secondary hyperparathyroidism of patients with early renal failure. *Am J Kidney Dis.* 1997;29:496–502.
- Gutierrez O, Isakova T, Rhee E, et al. Fibroblast growth factor-23 mitigates hyperphosphatemia but accentuates calcitriol deficiency in chronic kidney disease. *J Am Soc Nephrol.* 2005;16:2205–2215.
- Centeno PP, Herberger A, Mun HC, et al. Phosphate acts directly on the calcium-sensing receptor to stimulate parathyroid hormone secretion. *Nat Commun.* 2019;10:4693.
- Almaden Y, Canalejo A, Hernandez A, et al. Direct effect of phosphorus on PTH secretion from whole rat parathyroid glands in vitro. *J Bone Miner Res.* 1996;11:970–976.
- Slatopolsky E, Finch J, Denda M, et al. Phosphorus restriction prevents parathyroid gland growth. High phosphorus directly stimulates PTH secretion in vitro. *J Clin Invest.* 1996;97:2534–2540.
- Denda M, Finch J, Slatopolsky E. Phosphorus accelerates the development of parathyroid hyperplasia and secondary hyperparathyroidism in rats with renal failure. *Am J Kidney Dis.* 1996;28:596–602.
- Naveh-Many T, Rahamimov R, Livni N, et al. Parathyroid cell proliferation in normal and chronic renal failure rats. The effects of calcium, phosphate, and vitamin D. *J Clin Invest.* 1995;96:1786–1793.
- Ben-Dov IZ, Galitzer H, Lavi-Moshayoff V, et al. The parathyroid is a target organ for FGF23 in rats. *J Clin Invest.* 2007;117:4003–4008.
- Krajsnik T, Bjorklund P, Marsell R, et al. Fibroblast growth factor-23 regulates parathyroid hormone and 1 α -hydroxylase expression in cultured bovine parathyroid cells. *J Endocrinol.* 2007;195:125–131.
- Gutierrez OM, Mannstadt M, Isakova T, et al. Fibroblast growth factor 23 and mortality among patients undergoing hemodialysis. *N Engl J Med.* 2008;359:584–592.
- Gonzalez EA, Sachdeva A, Oliver DA, Martin KJ. Vitamin D insufficiency and deficiency in chronic kidney disease. A single center observational study. *Am J Nephrol.* 2004;24:503–510.
- Wada M, Furuya Y, Sakiyama J, et al. The calcimimetic compound NPS R-568 suppresses parathyroid cell proliferation in rats with renal insufficiency. Control of parathyroid cell growth via a calcium receptor. *J Clin Invest.* 1997;100:2977–2983.
- Brown EM, Gamba G, Riccardi D, et al. Cloning and characterization of an extracellular Ca(2+)-sensing receptor from bovine parathyroid. *Nature.* 1993;366:575–580.
- Murray TM, Rao LG, Divieti P, Bringham FR. Parathyroid hormone secretion and action: evidence for discrete receptors for the carboxyl-terminal region and related biological actions of carboxyl-terminal ligands. *Endocr Rev.* 2005;26:78–113.
- Slatopolsky E, Finch J, Clay P, et al. A novel mechanism for skeletal resistance in uremia. *Kidney Int.* 2000;58:753–761.
- Coen G, Bonucci E, Ballanti P, et al. PTH 1–84 and PTH “7–84” in the noninvasive diagnosis of renal bone disease. *Am J Kidney Dis.* 2002;40:348–354.
- Martin KJ, Akhtar I, Gonzalez EA. Parathyroid hormone: new assays, new receptors. *Semin Nephrol.* 2004;24:3–9.
- Monier-Faugere MC, Geng Z, Mawad H, et al. Improved assessment of bone turnover by the PTH-(1–84)/large C-PTH fragments ratio in ESRD patients. *Kidney Int.* 2001;60:1460–1468.
- Souberbielle JC, Boutten A, Carlier MC, et al. Inter-method variability in PTH measurement: implication for the care of CKD patients. *Kidney Int.* 2006;70:345–350.
- Malluche HH, Monier-Faugere MC. The role of bone biopsy in the management of patients with renal osteodystrophy. *J Am Soc Nephrol.* 1994;4:1631–1642.
- KDIGO 2017 clinical practice guideline update for the diagnosis, evaluation, prevention, and treatment of Chronic Kidney Disease–Mineral and Bone Disorder (CKD-MBD). *Kidney Int Suppl.* (2011). 2017; 7:1–59.
- Floege J. Phosphate binders in chronic kidney disease: a systematic review of recent data. *J Nephrol.* 2016;29:329–340.
- Toussaint ND, Lau KK, Polkinghorne KR, Kerr PG. Attenuation of aortic calcification with lanthanum carbonate versus calcium-based phosphate binders in haemodialysis: a pilot randomized controlled trial. *Nephrology (Carlton).* 2011;16:290–298.
- Kovesdy CP, Ahmadzadeh S, Anderson JE, Kalantar-Zadeh K. Association of activated vitamin D treatment and mortality in chronic kidney disease. *Arch Intern Med.* 2008;168:397–403.
- Shoben AB, Rudser KD, de Boer IH, Young B, Kestenbaum B. Association of oral calcitriol with improved survival in nondialyzed CKD. *J Am Soc Nephrol.* 2008;19:1613–1619.
- Investigators JD, Shoji T, Inaba M, et al. Effect of oral alfacalcidol on clinical outcomes in patients without secondary hyperparathyroidism receiving maintenance hemodialysis: the J-DAVID Randomized Clinical Trial. *JAMA.* 2018;320:2325–2334.
- Moe SM, Chertow GM, Coburn JW, et al. Achieving NKF-K/DOQI bone metabolism and disease treatment goals with cinacalcet HCl. *Kidney Int.* 2005;67:760–771.
- Raggi P, Chertow GM, Torres PU, et al. The ADVANCE study: a randomized study to evaluate the effects of cinacalcet plus low-dose vitamin D on vascular calcification in patients on hemodialysis. *Nephrol Dial Transplant.* 2011;26:1327–1339.
- Investigators ET, Chertow GM, Block GA, et al. Effect of cinacalcet on cardiovascular disease in patients undergoing dialysis. *N Engl J Med.* 2012;367:2482–2494.
- Block GA, Bushinsky DA, Cheng S, et al. Effect of etelcalcetide vs cinacalcet on serum parathyroid hormone in patients receiving hemodialysis with secondary hyperparathyroidism: a randomized clinical trial. *JAMA.* 2017;317:156–164.
- London GM, Marty C, Marchais SJ, et al. Arterial calcifications and bone histomorphometry in end-stage renal disease. *J Am Soc Nephrol.* 2004;15:1943–1951.
- Malluche HH, Mawad H, Monier-Faugere MC. The importance of bone health in end-stage renal disease: out of the frying pan, into the fire? *Nephrol Dial Transplant.* 2004;19(suppl 1):i9–i13.
- Behets GJ, Spasovski G, Sterling LR, et al. Bone histomorphometry before and after long-term treatment with cinacalcet in dialysis patients with secondary hyperparathyroidism. *Kidney Int.* 2015;87:846–856.
- Barreto FC, Barreto DV, Moyses RM, et al. K/DOQI-recommended intact PTH levels do not prevent low-turnover bone disease in hemodialysis patients. *Kidney Int.* 2008;73:771–777.
- Ferreira A, Frazao JM, Monier-Faugere MC, et al. Effects of sevelamer hydrochloride and calcium carbonate on renal osteodystrophy in hemodialysis patients. *J Am Soc Nephrol.* 2008;19:405–412.
- Sprague SM, Bellorin-Font E, Jorgetti V, et al. Diagnostic accuracy of bone turnover markers and bone histology in patients with CKD treated by dialysis. *Am J Kidney Dis.* 2016;67:559–566.
- Ok E, Asci G, Bayraktaroglu S, et al. Reduction of dialysate calcium level reduces progression of coronary artery calcification and improves low bone turnover in patients on hemodialysis. *J Am Soc Nephrol.* 2016;27:2475–2486.
- D’Haese PC, Spasovski GB, Sikole A, et al. A multicenter study on the effects of lanthanum carbonate (Fosrenol) and calcium carbonate on renal bone disease in dialysis patients. *Kidney Int Suppl.* 2003:S73–S78.
- London GM, Marchais SJ, Guerin AP, et al. Association of bone activity, calcium load, aortic stiffness, and calcifications in ESRD. *J Am Soc Nephrol.* 2008;19:1827–1835.
- Brandenburg VM, Floege J. Adynamic bone disease—bone and beyond. *NDT Plus.* 2008;1:135–147.

44. Sumida K, Ubara Y, Hoshino J, et al. Once-weekly teriparatide in hemodialysis patients with hypoparathyroidism and low bone mass: a prospective study. *Osteoporos Int*. 2016;27:1441–1450.
45. Peugh J, Khalil A, Chan MR, Hansen KE. Teriparatide treatment for hypercalcemia associated with adynamic bone disease. *JBMR Plus*. 2019;3:e10176.
46. Rubin MR, Zhou H, Cusano NE, et al. The effects of long-term administration of rhPTH(1-84) in hypoparathyroidism by bone histomorphometry. *J Bone Miner Res*. 2018;33:1931–1939.
47. Evenepoel P, Cunningham J, Ferrari S, et al. European Consensus Statement on the diagnosis and management of osteoporosis in chronic kidney disease stages G4-G5D. *Nephrol Dial Transplant*. 2021;36:42–59.
48. Jamal SA, West SL, Miller PD. Fracture risk assessment in patients with chronic kidney disease. *Osteoporos Int*. 2012;23:1191–1198.
49. Ballanti P, Wedard BM, Bonucci E. Frequency of adynamic bone disease and aluminum storage in Italian uraemic patients-retrospective analysis of 1429 iliac crest biopsies. *Nephrol Dial Transplant*. 1996;11:663–667.
50. Ishani A, Blackwell T, Jamal SA, et al. The effect of raloxifene treatment in postmenopausal women with CKD. *J Am Soc Nephrol*. 2008;19:1430–1438.
51. Jamal SA, Bauer DC, Ensrud KE, et al. Alendronate treatment in women with normal to severely impaired renal function: an analysis of the fracture intervention trial. *J Bone Miner Res*. 2007;22:503–508.
52. Miller PD, Roux C, Boonen S, et al. Safety and efficacy of risidronate in patients with age-related reduced renal function as estimated by the Cockcroft and Gault method: a pooled analysis of nine clinical trials. *J Bone Miner Res*. 2005;20:2105–2115.
53. Neer RM, Arnaud CD, Zanchetta JR, et al. Effect of parathyroid hormone (1-34) on fractures and bone mineral density in postmenopausal women with osteoporosis. *N Engl J Med*. 2001;344:1434–1441.
54. Jamal SA, Ljunggren O, Stehman-Breen C, et al. Effects of denosumab on fracture and bone mineral density by level of kidney function. *J Bone Miner Res*. 2011;26:1829–1835.
55. Floege J, Ketteler M. beta2-microglobulin-derived amyloidosis: an update. *Kidney Int Suppl*. 2001;78:S164–S171.
56. Zhang P, Fu X, Sawashita J, et al. Mouse model to study human A beta2M amyloidosis: generation of a transgenic mouse with excessive expression of human beta2-microglobulin. *Amyloid*. 2010;17:50–62.
57. Corlin DB, Heegaard NH. beta(2)-microglobulin amyloidosis. *Subcell Biochem*. 2012;65:517–540.
58. Jadoul M, Garbar C, Noel H, et al. Histological prevalence of beta 2-microglobulin amyloidosis in hemodialysis: a prospective post-mortem study. *Kidney Int*. 1997;51:1928–1932.
59. van Ypersele de Strihou C, Jadoul M, Malghem J, Maldague B, Jamart J. Effect of dialysis membrane and patient's age on signs of dialysis-related amyloidosis. The Working Party on Dialysis Amyloidosis. *Kidney Int*. 1991;39:1012–1019.
60. Ketteler M, Koch KM, Floege J. Imaging techniques in the diagnosis of dialysis-related amyloidosis. *Semin Dial*. 2001;14:90–93.